ModelIt!

Teacher's Guide

Ocean Currents and Climate: The Conveyor Belt

The Hidden Rivers That Control Our Weather

Quick Reference

Grade Level	Grade 07L2
NGSS Standard	MS-ESS2-6, MS-ESS3-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS2-6, MS-ESS3-5

Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates. Ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop and use a model to describe how unequal heating creates patterns of oceanic circulation that determine regional climates.
Disciplinary Core Idea	**ESS2.C The Roles of Water in Earth's Surface Processes / ESS3.D Global Climate Change** — Water's movements are driven by energy from the Sun and gravity, creating patterns that distribute heat and affect global climate.
Crosscutting Concept	**Energy and Matter** — Students trace the flow of thermal energy through the ocean current system, showing how energy transfer from equator to poles drives global climate patterns.

CAST Testing Alignment

MULTIPLE CHOICE:

London, England, and Goose Bay, Canada, are at nearly the same latitude. Yet London's average winter temperature is about 10 degrees Celsius warmer. Which factor best explains this difference?

CONSTRUCTED RESPONSE:

Using your ocean circulation model, explain how temperature differences between the equator and poles drive ocean currents and why disrupting these currents could change climate in regions far from where the disruption occurs.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The ocean covers 71% of Earth's surface and plays a critical role in regulating global climate. Ocean currents act as a massive heat transport system, moving thermal energy from tropical regions toward the poles and influencing weather patterns across entire continents.

The Great Ocean Conveyor Belt

The thermohaline circulation is a global system of deep-water currents driven by differences in temperature and salinity. Warm, salty surface water flows poleward (like the Gulf Stream), cools, becomes denser, and sinks to the ocean floor. This cold, dense water then flows back toward the equator along the ocean bottom, creating a continuous loop that takes about 1,000 years to complete.

Surface Currents and Wind

Surface currents are driven by wind patterns and the Coriolis Effect (Earth's rotation deflecting moving water). Trade winds push water westward near the equator, while prevailing westerlies push it eastward at mid-latitudes. These create circular current patterns called gyres that transport heat across ocean basins.

Ocean Currents and Climate

The Gulf Stream carries an enormous amount of heat from the tropical Atlantic to northwestern Europe. Without it, England would have winters similar to Labrador, Canada. Coastal cities near warm currents have milder climates, while those near cold currents (like San Francisco's fog from the California Current) have cooler summers.

Climate Change and Current Disruption

As global temperatures rise, polar ice melts and adds fresh water to the ocean. This reduces salinity, which could weaken the thermohaline circulation. Scientists monitoring the Atlantic Meridional Overturning Circulation (AMOC) have found it has slowed by about 15% since the mid-20th century, raising concerns about future climate disruption.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify temperature difference, wind patterns, ocean current speed, heat distribution, and regional climate stability as components of the ocean circulation system.

E **Establish Relationships**
Students determine that temperature differences drive current speed, which controls heat distribution, which in turn affects climate stability through cascading relationships.

V **Visualize & Model**
Students build a model showing how ocean circulation distributes thermal energy globally and stabilizes regional climates.

E **Evaluate Outcomes**
Students run normal circulation, enhanced gradient, and disrupted current scenarios to compare climate outcomes across regions.

R **Revise & Extend**
Students add salinity levels or Coriolis Effect to explore how additional factors influence ocean current behavior.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"There are rivers in the ocean. You can't see them, but they control your weather."*

Watch For

Show a NASA animation of ocean current patterns if available.

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling the planet's hidden heating system: ocean currents."*

Watch For

Have students read objectives. Pre-teach 'thermohaline circulation.'

Slide 3: Big Question

3 min

Say: *"London and northern Canada are at the same latitude. But London's winters are 10 degrees warmer. How?"*

Watch For

Show a map comparing London and Goose Bay. Discuss latitude vs. climate.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model of the ocean's heat delivery system."*

Watch For

Preview LEVER steps. Connect to what they know about hot and cold water.

Slide 5: Activity 1: Components

8 min

Say: *"What drives ocean currents? What can we control in our model vs. what responds?"*

Watch For

Guide sorting. Discuss why temperature difference and wind are external inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"When the temperature difference between equator and poles gets bigger, what happens to current speed? And how does that affect climate?"*

Watch For

Help students trace the cascade from temperature to currents to climate.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's see what happens when ocean currents are flowing normally vs. when they slow down."*

Watch For

Students predict first, then run simulations. Compare normal vs. disrupted scenarios.

Slide 8: Discoveries

5 min

Say: *"So WHY is Europe warmer than it should be? And what happens if we break the conveyor belt?"*

Watch For

Lead discussion about Gulf Stream. Show AMOC slowdown data.

Slide 9: STEM Challenge

12 min

Say: *"You're climate scientists now. Map what happens if ocean currents weaken by 50%."*

Watch For

Groups create impact maps using model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Temperature Difference (Equator vs Poles), Wind Patterns
- **INTERNAL:** Ocean Current Speed, Heat Distribution, Regional Climate Stability

Relationships

- **Temperature Difference to Ocean Current Speed:** POSITIVE (+) — A larger temperature gradient between the equator and poles creates a stronger density difference in ocean water, driving faster circulation through the conveyor belt.
- **Wind Patterns to Ocean Current Speed:** POSITIVE (+) — Stronger surface winds push more water, increasing the speed of surface currents and contributing to overall ocean circulation.
- **Ocean Current Speed to Heat Distribution:** POSITIVE (+) — Faster ocean currents transport more thermal energy from the tropics to the poles, creating more even heat distribution across the planet.
- **Heat Distribution to Regional Climate Stability:** POSITIVE (+) — When heat is distributed more evenly, coastal regions experience more moderate temperatures and more predictable weather patterns.
- **Heat Distribution to Temperature Difference:** NEGATIVE (-) — As currents distribute heat from equator to poles, they reduce the very temperature difference that drives them. This creates a self-regulating system.

Key Discoveries

- Ocean currents act as a global conveyor belt, carrying warm water toward the poles and cold water back toward the equator
- Wind patterns drive surface currents while temperature and salinity differences drive deep-water currents
- Europe is warmer than expected for its latitude because the Gulf Stream carries tropical heat northward
- If global warming disrupts the temperature gradient by melting polar ice, ocean currents could weaken, paradoxically making some regions colder

Multiple Choice Post-Assessment Answer Key

1. London, England, and Goose Bay, Canada, are at nearly the same latitude, yet London's average winter temperature is about 10 degrees Celsius warmer. Based on the ocean circulation model, which factor best explains this difference?

Correct: B) The Gulf Stream carries warm tropical water across the Atlantic to northwestern Europe, delivering heat that moderates London's climate

Correct! The Gulf Stream transports enormous amounts of thermal energy from the tropical Atlantic to northwestern Europe. This warm current heats the air above it, giving coastal Europe much milder winters than locations at the same latitude without this ocean heat delivery.

2. In the model, reducing the Temperature Difference between the equator and poles (simulating polar ice melt) causes Ocean Current Speed to decrease. Which cascade of effects does this trigger?

Correct: B) Slower currents reduce Heat Distribution, causing tropical regions to overheat while polar-adjacent regions lose warmth, destabilizing regional climates

Correct! When ocean currents slow, they transport less heat from the equator toward the poles. Heat accumulates in tropical regions while areas that depended on ocean heat delivery (like Europe) lose their warming input. This uneven distribution destabilizes regional climates.

3. A student claims that global warming should make ALL regions warmer, so Europe cannot get colder. Using the ocean circulation model, which argument best refutes this claim?

Correct: B) Global warming melts polar ice, reducing salinity and the temperature gradient that drives ocean currents. Weakened currents reduce heat delivery to Europe, potentially making it colder even as global averages rise

Correct! This counterintuitive outcome makes sense through the model. Global warming melts ice, adding fresh water that reduces ocean salinity and weakens the thermohaline circulation. Without the Gulf Stream delivering tropical heat, Europe could experience dramatically colder winters even as the global average temperature rises.

4. The model shows that Heat Distribution has a negative relationship with Temperature Difference. This means that as ocean currents distribute heat more evenly, the temperature gradient that drives them weakens. What type of mechanism does this represent?

Correct: B) A self-regulating negative feedback loop where the system moderates its own driving force

Correct! This is a self-regulating negative feedback loop. Ocean currents distribute heat from equator to poles, reducing the very temperature gradient that drives them. As the gradient weakens, currents slow, heat distribution becomes less even, and the gradient rebuilds. This keeps the system in dynamic balance.

5. In the model, both Temperature Difference and Wind Patterns have positive relationships with Ocean Current Speed. A student wants to determine which factor has a greater influence on current speed. Which investigation approach would best answer this question?

Correct: B) Change one variable at a time while holding the other constant, then compare the resulting changes in Ocean Current Speed

Correct! Controlled variable testing is the proper scientific approach. By changing one input while holding the other constant, the student can isolate each factor's individual effect on current speed and determine which has a greater influence.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The ocean is still water"	The ocean is constantly in motion. Surface currents driven by wind can move water at speeds up to 5 mph, while deep-water thermohaline currents move more slowly but continuously. The Great Ocean Conveyor Belt circulates water through the entire global ocean over about 1,000 years.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Ocean currents only affect coastal areas"	Ocean currents affect climate far inland by warming or cooling the air that blows over them. The Gulf Stream warms air masses that then blow across all of Western Europe, affecting temperatures hundreds of miles from the coast. Ocean currents also drive evaporation that creates rain patterns over entire continents.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Climate and weather are the same"	Weather is what happens today or this week (sunny, rainy, cold snap). Climate is the average pattern over 30+ years. Ocean currents affect CLIMATE by consistently delivering heat to certain regions. A single cold day doesn't change the climate, but a persistent change in ocean currents would.